

Milestone 3 – Final Report

DSC 680 - Applied Data Science

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Executive Summary

Studios and publishers require a reliable method for estimating expected sales performance before release. Production budgets, marketing allocations, and platform commitments depend on accurate forecasting. Historical sales data provides labeled outcomes; modern Steam metadata establishes how current titles are positioned in the market. The objective is to identify the factors that drive sales and build a baseline model capable of forecasting performance for new or existing games using features shared between historical and modern datasets. A Random Forest Regressor trained on Platform, Genre, Publisher, and Year achieved a mean squared error (MSE) of 0.1313 on log-transformed global sales. Feature importance results confirm that publisher identity, genre, and release year are the strongest predictors. This white paper documents the data sources, preparation workflow, modeling approach, results, limitations, and a complete question-and-answer section addressing ten audience questions.

1. Introduction

Forecasting video game sales before release is a core requirement for planning and resource allocation. Historical sales patterns show that platform, genre, publisher, and release year consistently influence performance. Steam metadata provides modern context but contains inconsistent fields across merged sources. Only four features overlap between the historical dataset and the Steam metadata: Platform, Genre, Publisher, and Year. These four shared fields form the operational baseline for a supervised regression model.

The goal is to train a model on labeled historical sales data and apply it to modern Steam titles to generate estimated global sales figures. Ethical considerations include potential bias in critic and user review scores and the use of SteamSpy owner estimates as proxies for actual sales. The analysis does not involve personal or demographic data and does not make claims about individuals. Focus remains on aggregate patterns and predictive performance.

2. Data Description

2.1 Historical Sales Dataset

The primary dataset is the "Video Game Sales with Ratings" dataset sourced from Kaggle. It contains 16,598 records and includes the following fields: Platform, Genre, Publisher, Year, and Global_Sales. Global_Sales serves as the target variable. Year, Platform, Genre, and Publisher function as the predictive features. The dataset spans multiple console generations and captures a wide range of genres and publishers. Regional sales columns are present but are not used in the baseline model.

2.2 Steam Metadata

Steam metadata was constructed by merging three datasets - steam_games, steam_history, and master_games - joined on appid/App_ID. The merged dataset includes positive and negative review counts, userscore, owner estimates expressed as range strings, price and discount information, concurrent user counts, playtime metrics, partial genre labels, and partial release dates. Only four fields align with the historical dataset: Platform (assigned as PC for all Steam titles), Genre, Publisher, and Year. These shared fields constitute the baseline feature set used for forecasting when applying the trained model to modern titles.

3. Data Preparation

3.1 Historical Sales Cleaning

Records with missing values in Year, Genre, Platform, or Publisher were removed. Titles with zero global sales were excluded. Year was cast to integer type. Global_Sales was log-transformed using log1p to reduce right skew and stabilize variance for regression. This process produced a clean, labeled dataset suitable for supervised learning.

3.2 Steam Metadata Merge and Cleaning

The three Steam datasets were merged on appid/App_ID using left joins. Release year was rebuilt from Release_Date_x, with missing values filled using Release_Date_y. The unified date field was converted to datetime format and the year extracted. Owner ranges reported as strings (e.g., "20000–50000") were converted to numeric midpoints using a custom midpoint function. Review ratio was computed as positive reviews divided by total reviews (positive plus negative). Genre was unified from Main_Genre_x and Main_Genre_y using fillna. Publisher was unified from publisher, Developer_x, and Developer_y using sequential fillna. Platform was assigned the

value "PC" for all Steam records. Unknown or missing Genre and Publisher values were filled with "Unknown."

3.3 Feature Alignment

A feature frame was constructed using only Platform, Genre, Publisher, and Year - the four fields shared between the historical and Steam datasets. Year was imputed using the column median due to incomplete release date coverage in the Steam data. Publisher_Final was renamed to Publisher to ensure column name consistency. This alignment guarantees that the model trained on historical data can be applied to Steam titles without feature mismatch or schema conflicts.

4. Modeling Approach

A supervised regression model was selected to predict log-transformed global sales. The four shared features - Platform, Genre, Publisher, and Year - serve as predictors. A Random Forest Regressor (random_state=42) was trained using an 80/20 train-test split. A preprocessing pipeline applied one-hot encoding (handle_unknown="ignore") to the three categorical features and passed Year through unchanged via passthrough. The pipeline structure ensures that unseen categories in the test set or Steam data do not cause prediction errors.

Model performance was evaluated using mean squared error on the held-out test set. The baseline model achieved an MSE of 0.1313. Enhanced models incorporating Steam-only features such as owners_mid, review_ratio, price, and concurrent user counts are identified as future work and are deferred to later milestones.

5. Results

5.1 Genre Effects

A boxplot of Global Sales by Genre reveals high variance in Action, Shooter, Sports, and Role-Playing titles. Puzzle, Strategy, and Misc genres show comparatively low variance. This pattern confirms that genre is a major driver of sales outcomes and supports its inclusion as a predictive feature.

5.2 Temporal Effects

A scatterplot of Global Sales by Year shows a concentration of high-selling titles in earlier console generations and a clear decline in peak sales figures after approximately 2010. Earlier years contain sparse but high-variance observations. This lifecycle pattern confirms that release year influences sales performance and justifies its inclusion as a model feature.

5.3 Platform Distribution

A bar chart of platform counts shows heavy concentration on DS, PS2, PS3, Wii, and X360. Many other platforms appear infrequently in the historical data. PC is present but is not a dominant contributor in the historical dataset. This imbalance supports treating Platform as a categorical predictor and explains why Steam-based predictions cluster around PC-specific historical patterns.

5.4 Feature Importance

Random Forest feature importance results identify Year, Publisher categories, and Genre categories as the dominant predictors. Platform contributes to predictive power but ranks lower than the other three features. These results directly address the research question by confirming that publisher identity, genre classification, and release timing are the strongest drivers of global sales in the historical dataset.

5.5 Model Fit

A scatterplot comparing actual and predicted log-transformed global sales on the test set (Figure 5) shows predictions following the diagonal trend with moderate spread. The model generalizes and produces usable baseline forecasts. The spread around the diagonal reflects normal variance for a model constrained to four shared features. Sample predictions on unique Steam titles ranged from a minimum of approximately 0.012 to a maximum of approximately 0.297 predicted global sales (in millions, post-`expm1` transformation). The mean predicted sales value across 9,950 unique Steam titles was 0.020, with a standard deviation of 0.025.

6. Limitations

Steam metadata contains incomplete release dates, requiring median imputation for Year. The owners field is a range estimate, not an exact count; the midpoint conversion introduces measurement error. Genre and Publisher fields are inconsistent across the three Steam source datasets and require heuristic unification. The historical dataset reflects sales patterns from earlier console generations and may not represent current market conditions.

The baseline model uses only four shared features; predictive power is bounded by this constraint. Enhanced models incorporating Steam-only variables - owners_mid, review_ratio, price, and concurrent user counts - require additional feature engineering and are deferred to later milestones. High-selling titles are underrepresented in the training data due to sales distribution skew, which may cause the model to underpredict outlier performance

7. Ten Audience Questions

1. How accurate is the model when predicting high-selling titles?

The baseline model achieved an MSE of 0.1313 on log-transformed global sales, indicating reasonable generalization across the test set. However, the sales distribution is heavily right-skewed, with the majority of titles recording low global sales and a small number reaching high-selling status. The model tends to underpredict high-selling outliers because they are underrepresented in training data relative to low-selling titles. The actual-versus-predicted scatterplot confirms that predictions follow the diagonal trend but exhibit increasing spread at higher sales values. Improving accuracy for high-selling titles would require oversampling or stratified training techniques targeted at the upper tail of the distribution.

2. How does genre influence sales across different platforms?

The boxplot of Global Sales by Genre shows that Action, Shooter, Sports, and Role-Playing titles exhibit the highest variance and the highest upper-end sales values. Puzzle, Strategy, and Misc genres consistently produce low-variance, lower-sales outcomes. Platform distribution in the historical dataset is dominated by DS, PS2, PS3, Wii, and X360, meaning the genre-platform interaction in training data reflects those console generations. Because Steam titles are all assigned the Platform value of "PC," genre effects in Steam predictions are filtered through PC-specific historical patterns. Genre remains the second most influential feature grouping behind Publisher in the Random Forest importance rankings.

3. How sensitive is the model to missing publisher information?

Publisher is encoded using one-hot encoding with the `handle_unknown="ignore"` setting, which means unrecognized publisher values during prediction produce zero-valued columns for all publisher dummies. This causes the model to route those records through non-publisher feature signals - Year, Genre, and Platform - exclusively. Publishers with sparse representation in the training set may behave similarly to unknown publishers due to limited training signal. Records where Publisher was missing were filled with "Unknown" prior to modeling, creating a dedicated category that captures residual patterns from unidentified publishers. Sensitivity to missing publisher data is a known limitation of the baseline approach and motivates future work on publisher embedding or frequency-based encoding.

4. How does the model handle new genres not present in the training data?

New genres not seen during training are handled by the `OneHotEncoder's` `handle_unknown="ignore"` parameter, which produces an all-zero encoding for that genre column. As a result, the model generates predictions based solely on the remaining features - Publisher, Platform, and Year - without any genre signal contributing to the output. This produces predictions that default toward the mean behavior of whatever publisher and year combination is present, which may not reflect the actual market behavior of a genuinely novel genre. The impact is expected to be limited in practice because the major commercial genres are well-represented in the historical dataset. Monitoring for new genre labels in incoming Steam data and retraining periodically would reduce this risk over time.

5. How reliable is the `owners_mid` field as a proxy for sales?

The owners field in Steam metadata is reported as a string range (e.g., "20,000–50,000") and converted to a numeric midpoint estimate. This introduces inherent measurement error because the actual owner count may fall anywhere within the range, and the ranges themselves are broad at higher ownership levels. SteamSpy owner estimates are acknowledged proxies rather than verified sales figures. Despite this limitation, owners_mid provides an ordinal signal that is directionally consistent with game popularity and is useful as an input to enhanced models. Its reliability as a direct sales proxy is low, but its value as a relative ranking feature remains meaningful for comparative forecasting.

6. What improvements occur when adding Steam-only features?

The baseline model uses only four shared features and achieves an MSE of 0.1313 on log-transformed sales. Adding Steam-only features such as owners_mid, review_ratio, price, and concurrent user counts would provide additional signal about modern market engagement that the historical dataset cannot supply. Review ratio captures community sentiment, owners_mid provides an ownership magnitude estimate, and concurrent users reflect active player engagement - all plausible predictors of sales performance. These features are identified as targets for enhanced model development in later milestones. Quantified improvement metrics are not yet available because the enhanced model has not been trained, but directional improvement is expected given the informational value of these fields.

7. How does release year affect prediction accuracy?

Year is the single highest-ranked feature in the Random Forest importance output, indicating that release timing carries more predictive weight than Platform and comparable weight to the top publisher and genre categories. The scatterplot of Global Sales by Year confirms a lifecycle pattern in which high-selling titles cluster in earlier console generations, with peak sales declining after approximately 2010. This means the model's predictions for recent Steam titles are influenced by the declining trend observed in post-2010 historical data, which may cause underestimation for currently successful franchises. Year was imputed using the median for Steam records with missing release dates, which introduces noise for titles where the actual release year would meaningfully shift the prediction. Maintaining accurate release year data in Steam inputs is the single highest-impact data quality action for improving model output.

8. How does the model perform across different publishers?

Publisher is the feature group with the most individual one-hot encoded columns and collectively contributes significant predictive power in the Random Forest importance rankings. Publishers with high historical sales volume and many training records receive more accurate predictions because the model has sufficient examples to learn their patterns. Publishers with few historical records or those appearing only in Steam data are predicted with lower confidence, defaulting toward genre and year signals. The actual-versus-predicted scatterplot does not break out error by publisher, but the skew in publisher representation means performance is uneven across the publisher distribution. Future analysis stratified by publisher size or category would identify where the model performs well and where additional training data would have the highest impact.

9. What are the main sources of error in the predictions?

Four primary sources of error are identified. First, sales distribution skew causes the model to underpredict high-selling titles because the training set is dominated by low-selling records. Second, feature sparsity limits predictive resolution - using only four features leaves substantial variance in global sales unexplained. Third, temporal mismatch between the historical dataset (older console generations) and current Steam titles means the model applies patterns from a different market era to modern predictions. Fourth, imputation of missing Year values using the median introduces systematic noise for Steam titles with incomplete or unavailable release date information. Reducing these error sources requires expanded feature sets, stratified training approaches, and updated historical data reflecting post-2015 market behavior.

10. How would this model integrate into a real studio workflow?

In a studio context, the model would function as a pre-release screening tool rather than a production sales guarantee. A title in development would be characterized by its target platform, primary genre, publishing entity, and planned release year; those four values would be passed through the trained pipeline to generate a predicted global sales estimate. The predicted output, expressed in millions of units after inverse log transformation, would inform green-light decisions, marketing budget allocations, and platform prioritization discussions. The model would require periodic retraining as new historical data becomes available to remain calibrated to current market conditions.

Integration with internal data warehouses and build-time metadata systems would allow the prediction to be generated automatically as part of the title planning process, reducing analyst time and increasing forecast consistency.

8. Appendix – Supporting Documentation

Include:

- Full code used for data cleaning, merging, feature engineering, and modeling
- Data dictionaries for vg_sales and Steam datasets
- Merge logic diagram
- Missingness summary before and after cleaning
- Feature alignment table

This appendix supports reproducibility and technical validation.

9. References (APA)

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